MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 03 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions
MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is: [1]

(A) $-5/4$ (B) $2/5$ (C) $15/4$ (D) $3/2$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C)** $15/4$ [0.5 Mark]

Rewriting as $3x + 2ky - 2 = 0$ and $2x + 5y + 1 = 0$. For parallel lines:

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Justification: [0.5 Mark]

$\frac{3}{2} = \frac{2k}{5} \implies 4k = 15 \implies k = \mathbf{\frac{15}{4}}$ (also $\frac{3}{2} \neq \frac{-2}{1}$ holds).

Q2. The point of intersection of the line represented by $3x - y = 3$ and the y-axis is: [1]

(A) $(0, -3)$ (B) $(0, 3)$ (C) $(1, 0)$ (D) $(-3, 0)$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** $(0, -3)$ [0.5 Mark]

Justification: Every point on the y-axis has equation $x = 0$. [0.5 Mark]

Substituting $x = 0$ into $3x - y = 3$: $3(0) - y = 3 \implies y = -3$. Hence point is $(\mathbf{0, -3})$.

Q3. The area of the triangle formed by the lines $y = x$, $x = 6$, and $y = 0$ is: [1]

(A) 36 sq. units (B) 18 sq. units (C) 9 sq. units (D) 72 sq. units

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 18 sq. units** [0.5 Mark]

Formula / Theorem Used: Area of right triangle $= \frac{1}{2} \times \text{Base} \times \text{Height}$

Justification: Vertices are intersections $(0, 0)$, $(6, 0)$, and $(6, 6)$. [0.5 Mark]

Base $= 6$ units, Height $= 6$ units $\implies$ Area $= \frac{1}{2} \times 6 \times 6 = \mathbf{18}$ sq. units.

SECTION B: Assertion-Reason Question — MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The system of equations $2x+y+9=0$ and $x+3y+7=0$ is consistent with a unique solution.
Reason (R): The system $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is consistent with a unique solution if $a_1b_2 \neq a_2b_1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Here $a_1b_2 = (2)(3) = 6$ and $a_2b_1 = (1)(1) = 1$, so $a_1b_2 \neq a_2b_1$ (equivalently $\frac{2}{1} \neq \frac{1}{3}$). [0.5 Mark]

Hence both (A) and (R) are true and (R) correctly explains (A).

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. Determine the value of k for which the given system of equations has no solution:

$$3x + y = 1 \quad \text{and} \quad (2k-1)x + (k-1)y = 2k+1$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Formula / Theorem Used: No Solution Condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Step 1: Equating Coefficient Ratios [1 Mark]

$$\frac{3}{2k-1} = \frac{1}{k-1} \neq \frac{1}{2k+1}$$

Step 2: Solving for k and Verifying Inequality [1 Mark]

$3(k-1) = 2k-1 \implies 3k-3 = 2k-1 \implies k = 2$.

At $k=2$: $\frac{1}{k-1} = 1 \neq \frac{1}{2(2)+1} = \frac{1}{5}$ (parallel condition holds).

$$\boxed{\mathbf{k = 2}}$$

Q6. Solve the following pair of linear equations using the elimination method:

$$3x + 4y = 10 \quad \text{and} \quad 2x - 2y = 2$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $3x + 4y = 10$ — (i) and $2x - 2y = 2$ — (ii)

Step 1: Eliminating y [1 Mark]

Multiplying equation (ii) by 2: $4x - 4y = 4$ — (iii)

Adding (i) and (iii): $(3x+4y) + (4x-4y) = 10+4 \implies 7x = 14 \implies x = 2$.

Step 2: Solving for y [1 Mark]

Substituting $x = 2$ into (ii): $2(2) - 2y = 2 \implies 4 - 2y = 2 \implies 2y = 2 \implies y = 1$.

$$\boxed{\mathbf{x = 2, \quad y = 1}}$$

Q7. The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number. How many such numbers are there? [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let the tens digit be x and units digit be y.

Formula / Theorem Used: Original Number $= 10x + y$, Reversed Number $= 10y + x$

Step 1: Equation from Sum of Numbers [1 Mark]

$$(10x + y) + (10y + x) = 66 \implies 11(x + y) = 66 \implies x + y = 6 \quad \text{— (i)}$$

Step 2: Considering Both Digit-Difference Cases [2 Marks]

Since digits differ by 2, either $x - y = 2$ or $y - x = 2$:

Case I $(x - y = 2)$**:** Adding to (i): $2x = 8 \implies x = 4, y = 2 \implies$ Number $= 42$.

Case II $(y - x = 2)$**:** Adding to (i): $2y = 8 \implies y = 4, x = 2 \implies$ Number $= 24$.

Numbers are **42** and **24**; There are **2** such numbers.

Q8. Draw the graphs of the linear equations $4x - y - 8 = 0$ and $2x - 3y + 6 = 0$. [5]

(i) Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis.

(ii) Determine the coordinates of the vertices of the triangle formed by these lines and the y-axis.

(iii) Calculate the ratio of the areas of the two triangles.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

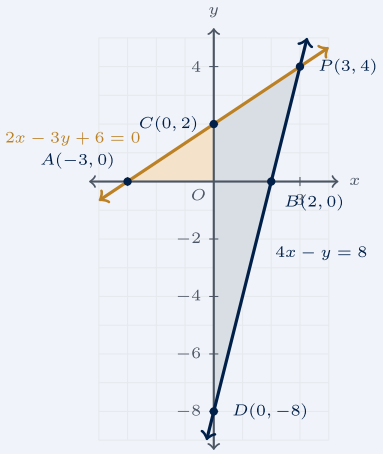
Step 1: Solution Tables [1.5 Marks]

$y = 4x - 8$ and $y = \frac{2x+6}{3}$:

x	0	2	3
$y = 4x - 8$	-8	0	4

x	-3	0	3
$y = \frac{2x+6}{3}$	0	2	4

Both lines intersect at $P(3, 4)$.



(i) Triangle with x-axis $(\triangle PAB)$: [1 Mark]

Vertices: **P(3, 4)**, **A(−3, 0)**, **B(2, 0)**.

Base $AB = 2 - (-3) = 5$, Height $= 4 \implies$ Area$_1 = \frac{1}{2}(5)(4) =$ **10** sq. u.

(ii) Triangle with y-axis $(\triangle PCD)$: [1 Mark]

Vertices: **P(3, 4)**, **C(0, 2)**, **D(0, −8)**.

Base $CD = 2 - (-8) = 10$, Height $= 3 \implies$ Area$_2 = \frac{1}{2}(10)(3) =$ **15** sq. u.

(iii) Ratio of Areas: [1.5 Marks]

$$\text{Area}_1 : \text{Area}_2 = 10 : 15 = \boxed{\mathbf{2 : 3}}$$

Q9. CASE STUDY: Social Welfare Coaching Centre Fee Collection

A non-profit educational society conducts remedial mathematics classes for students from diverse backgrounds. To support operations, children from affluent families pay a regular fee of Rs. y per month, while children from economically weaker sections (EWS) receive financial assistance and pay a nominal subsidized fee of Rs. x per month. In Batch I, there are 20 EWS children and 5 regular children, and the total fee collected is Rs. 9000. In Batch II, there are 5 EWS children and 25 regular children, and the total fee collected is Rs. 26000. [4]

(i) Represent the fee structure algebraically as a pair of linear equations in x and y. [1]

(ii) Find the monthly fee paid by an EWS child (x). [1]

(iii) Find the monthly fee paid by a regular child (y). [1]

(iv) If Batch III consists of 10 EWS children and 20 regular children, calculate the total collection. [1]

— OR —

Find the difference in monthly fees paid by a regular child and an EWS child. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Formula / Theorem Used: Total Fee Collection = (EWS Count)x + (Regular Count)y

Sub-part (i): Algebraic Formulation [1 Mark]

Batch I: $20x + 5y = 9000 \implies 4x + y = 1800$ — (i)

Batch II: $5x + 25y = 26000 \implies x + 5y = 5200$ — (ii)

Sub-part (ii): Monthly Fee for an EWS Child (x) [1 Mark]

From (i), $y = 1800 - 4x$. Substituting into (ii):

$$x + 5(1800 - 4x) = 5200 \implies x + 9000 - 20x = 5200 \implies -19x = -3800 \implies \boxed{\mathbf{x = \text{Rs. } 200}}$$

Sub-part (iii): Monthly Fee for a Regular Child (y) [1 Mark]

$y = 1800 - 4(200) = 1800 - 800 = \boxed{\mathbf{y = \text{Rs. } 1{,}000}}$.

Sub-part (iv): Total Collection for Batch III (10 EWS, 20 Regular) [1 Mark]

Total $= 10x + 20y = 10(200) + 20(1000) = 2000 + 20000 = \boxed{\text{Rs. } \mathbf{22{,}000}}$.

Sub-part (iv) [OR Option]: Difference in Monthly Fees [1 Mark]

Difference $= y - x = 1000 - 200 = \boxed{\text{Rs. } \mathbf{800}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★